4.5.1 Proportion Method

In this method, we shall compare the given data and find whether they are in direct or inverse proportion. By finding the proportion, we can use the fact that

the product of the extremes = the product of the means

and get the value of the unknown (x).

4.5.2 Multiplicative Factor Method

Illustration:

Men	Hours	Days
D $\begin{matrix} a \\ x \end{matrix}$ I	D $\begin{matrix} c \\ d \end{matrix}$	$\begin{matrix} e \\ f \end{matrix}$ I

Here, the unknown (x) is in men and so it is compared to the known, namely hours and days. If men and hours are in direct proportion (D), then take the multiplying factor as $\frac{d}{c}$ (take the reciprocal). Also, if men and days are in inverse proportion (I), then take the multiplying factor as $\frac{e}{f}$ (no change). Thus, we can find the unknown (x) in men as $x = a \times \frac{d}{c} \times \frac{e}{f}$.

4.5.3 Formula Method

Identify the data from the given statement as Persons (P), Days (D), Hours (H) and Work (W) and use the formula,

$$\frac{P_1 \times D_1 \times H_1}{W_1} = \frac{P_2 \times D_2 \times H_2}{W_2}$$

where the suffix 1 contains the complete data from the first statement of the given problem and the suffix 2 contains the unknown data to be found out in the second statement of the problem. That is, this formula says, P_1 men doing W_1 units of work in D_1 days working H_1 hours per day is the same as P_2 men doing W_2 units of work in D_2 days working H_2 hours per day. Identifying the work W_1 and W_2 correctly is more important in these problems. This method will be easy for finding the unknown (x) quickly.

Example 4.19 **(Direct – Direct Variation)**

If a company pays ₹6 lakh for 15 workers for 20 days, how much would it need to pay for 5 workers for 12 days?

Solution:

Proportion Method

Workers	Pay (Work)	Days
D $\begin{matrix} 15 \\ 5 \end{matrix}$	D $\begin{matrix} 6 \\ x \end{matrix}$ D	$\begin{matrix} 20 \\ 12 \end{matrix}$ D

Here, the unknown is the pay (x). It is to be compared with the workers and the days.

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